THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. If 1 lakh = 1,00,000, how many lakhs are there in 1,25,00,000? 1

(a) 126 **(b)** 125 **(c)** 124 **(d)** 135

2. A cyclist covers 30 km in the first hour and 2 km less every following hour for 5 hours in all. The total distance (km) is: 1

(a) 135 **(b)** 129 **(c)** 131 **(d)** 130

3. What must be added to 2.35 to get 5? 1

(a) 3.65 **(b)** 2.75 **(c)** 2.55 **(d)** 2.65

4. If $x = 4$ and $y = 5$, the value of $3x - 2y + xy$ is: 1

(a) 21 **(b)** 24 **(c)** 22 **(d)** 23

5. Two angles of a linear pair differ by $20°$. The smaller angle is: 1

(a) $100°$ **(b)** $80°$ **(c)** $70°$ **(d)** $90°$

6. If $1 + 2 + 3 + \cdots + n = 231$, then n is: 1

(a) 21 **(b)** 23 **(c)** 22 **(d)** 20

7. In an isosceles triangle with perimeter 34 cm the equal sides are 12 cm each. The base (cm) is: 1

(a) 10 **(b)** 9 **(c)** 12 **(d)** 11

8. $\frac{3}{5}$ of $3\frac{1}{3}$ is: 1

(a) $3\frac{14}{15}$ **(b)** 4 **(c)** 1 **(d)** 2

9. The number of digits in the product 1,25,000 × 8,000 is: 1

(a) 11 **(b)** 10 **(c)** 9 **(d)** 12

10. The value of $(3 \times (18 - 4) + 6) \div (2 \times (7 + 1))$ is: 1

(a) 197 **(b)** 3 **(c)** 72 **(d)** 5

11. Which fraction is equal to 0.125? 1

(a) $\frac{1}{125}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{1}{12}$ **(d)** $\frac{1}{4}$

12. A rule is: output $= 3\times$ input -4. The output for the input 9 is: 1

(a) 24 **(b)** 23 **(c)** 25 **(d)** 22

13. One angle of a linear pair is 2 times the other. The larger angle is: 1

(a) $140°$ **(b)** $120°$ **(c)** $110°$ **(d)** $130°$

14. The number of reverse-and-add steps for 59 to become a palindrome is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

15. The two acute angles of a right triangle are in the ratio $1:2$. The smaller acute angle is: 1

(a) 20° **(b)** 40° **(c)** 50° **(d)** 30°

16. If $x \times \frac{3}{4} = \frac{9}{16}$, then $x \div \frac{3}{4}$ is: 1

(a) $1\frac{1}{2}$ **(b)** 2 **(c)** $\frac{1}{2}$ **(d)** 1

17. **Assertion (A):** 1 million = 10 lakh. 1
Reason (R): 1 billion = 1000 million.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** $(3+4)\times 2 = 14$. 1
Reason (R): $3 + 4 \times 2 = 11$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** 2.50 = 2.5. 1
Reason (R): 0.1 = 1/10.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** The value of 2x + 3 when x = 4 is 11. 1
Reason (R): In 2x, the number 2 and the letter x are multiplied.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. An angle is 6° more than 3 times its complement. Find the angle. 2

22. Show that $n(n+1)$ is even, and verify for $n = 13$. 2

23. In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 115^\circ$. Find $\angle A$ and $\angle B$. 2

24. After $\frac{2}{7}$ of a number is removed, 35 is left. Find the number and the part removed. 2

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25. A number is 1 lakh more than 1,23,45,678. Write the number and the number which is 1 crore less than it (if possible). 2

26. Without finding the products, compare 15 × 10 and 14 × 11. Check by computing. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. The speed of light is about 3,00,000 km per second. How far does it travel in 8 seconds? Write the answer in words. 3

28. A rectangular garden is 20 m long and 12 m wide. A path of width 2 m runs around outside. Write an expression for the area of the path and evaluate it. 3

29. 2.5 kg of rice costs ₹135. Find the price of 1 kg and the cost of 4.5 kg. 3

30. Taxi A charges ₹20 plus ₹15 per km and taxi B charges ₹60 plus ₹10 per km. Write both fares for k km. For a journey of 6 km, which taxi is cheaper and by how much? 3

31. Two adjacent angles form a linear pair and one of them is 40°. Find the other angle, the half of each angle and the angle between the two bisectors. 3

OR

In how many ways can 15 be written as the sum of two or more consecutive positive whole numbers? List them. 3

32. Two sides of a triangle are 8 cm and 21 cm. The third side is a whole number of cm divisible by 5. List all possible lengths and the largest perimeter. 3

OR

A pole of 20 m is cut into pieces of $1\frac{2}{3}$ m each. How many pieces are obtained and how much is left? 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Two railway tracks are parallel. A road crosses them and acts as a transversal. At the first crossing, one of the angles is 65°. 5

(i) What is the corresponding angle at the second crossing?

(ii) What is the alternate interior angle?

(iii) What is the angle on the same side of the transversal, between the tracks (same-side interior angle)?

(iv) Of the eight angles formed, how many measure 65° and how many measure 115°?

34. **Case study.** Teacher Neha shows how 15 can be written as a sum of consecutive numbers: $15 = 7 + 8 = 4 + 5 + 6 = 1 + 2 + 3 + 4 + 5$. 5

(i) Which of these sums uses exactly three numbers?

(ii) Write 21 as the sum of three consecutive numbers.

(iii) Is the sum of three consecutive numbers always divisible by 3? Give the reason.

(iv) Write 33 as the sum of three consecutive numbers and also as the sum of two consecutive numbers.

35. Three angles on one side of a straight line at a point are in the ratio $2 : 3 : 4$. Find the three angles and name the kind of each. 5

36. Use reverse-and-add on 155 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

OR

The exterior angles of a triangle (one at each vertex) are in the ratio $4 : 5 : 6$. Find the three exterior angles, the three interior angles and classify the triangle. 5

37. A farmer's land is 360 acres. He sells $\frac{2}{5}$ of it, gives $\frac{1}{6}$ of the remainder to his son and plants $\frac{4}{5}$ of what is left after that with sugarcane. Find each area and the area left unplanted. 5

OR

Two numbers add up to 98,00,000 and differ by 24,00,000. Find the numbers, and write both in words. 5

38. Evaluate $(20 + 4) \times 5$ and $20 + (4 \times 5)$. Which bracketing gives the larger value and by how much? 5

OR

A shopkeeper mixes 2 kg of tea costing ₹240 per kg with 3 kg of tea costing ₹180 per kg. Find the total cost, the total weight and the cost of 1 kg of the mixture. 5

END OF QUESTION PAPER **Total: 80 Marks**

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